

Replication of Table 1 from Arkhangelsky et al. (2021)

Lauren Taylor

June 4, 2026

1 The Original Paper

Arkhangelsky, Athey, Hirshberg, Imbens, and Wager (2021) introduce *Synthetic Difference-in-Differences* (SDID), a method for estimating the causal effect of a policy when only a few units are treated (Arkhangelsky et al., 2021). Their application is California’s Proposition 99, a 1989 cigarette tax, and they pose the question: how much did it reduce smoking?

Two existing methods each have a known weakness. Difference-in-Differences (DID) compares treated and control units before and after the policy, but requires that both groups would have followed parallel trends without the intervention, which is an assumption that often fails in practice. Synthetic Control (SC) builds a weighted average of control states that matches the treated state’s pre-policy trend, but provides no standard errors and can be sensitive to which control states are included. SDID combines both: it uses SC-style weights to make the comparison group look like California before 1989, then applies DID to estimate the treatment effect. The result is an estimator with better balance than DID and the formal inference that SC lacks.

2 The Result of Interest

I replicated **Table 1** of the paper, which reports the estimated effect of Proposition 99 on annual per-capita cigarette sales under five methods: SDID, SC, DID, DIFP, and MC. The paper’s targets are -15.6 packs (SDID), -19.6 (SC), -27.3 (DID), -11.1 (DIFP), and -20.2 (MC), with placebo standard errors alongside each.

Table 1 is the paper’s central result as it compares all these estimation styles. Reproducing it requires implementing the Frank-Wolfe weight-estimation algorithm from scratch in Python, since no official updated Python package exists.

3 Data and Methods

Data

The data come from a panel of 39 U.S. states observed annually from 1970 to 2000, originally compiled by Abadie et al. (2010) and distributed with the authors’ replication package (Arkhangelsky et al., 2021). The outcome is per-capita cigarette sales (packs per year) and California is the only treated state (Prop 99 started in 1989); the other 38 states serve as controls.

`code/preprocess.py` loads the raw CSV and writes a clean state-by-year panel matrix to `temp/panel.csv`.

Estimators

`code/analysis.py` implements four of the five estimators. All share the same panel dimensions: $N = 39$ states, $T = 31$ years, $N_0 = 38$ controls, $T_0 = 19$ pre-treatment years.

SDID. The estimator finds weights $\hat{\omega}$ (one per control state) and $\hat{\lambda}$ (one per pre-treatment year) such that the weighted control average tracks California’s pre-1989 trend as closely as possible. It then runs a weighted two-way fixed-effects regression:

$$\hat{\tau}^{\text{sdid}} = \arg \min_{\tau, \mu, \alpha, \beta} \sum_{i,t} (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2 \hat{\omega}_i \hat{\lambda}_t \quad (1)$$

Both weight sets are estimated by Frank-Wolfe optimization which is a simple iterative algorithm that moves weight onto whichever state or year best reduces the fitting error at each step.

SC. Like SDID but with no time weights: it finds the weighted combination of control states that best matches California’s entire pre-treatment history, then compares post-treatment averages directly.

DID. All states and all years receive equal weight.

DIFP. A hybrid proposed by Doudchenko and Imbens (2016): SC-style unit weights (with an intercept) combined with uniform time weights.

MC. The matrix completion estimator of Athey et al. (2021) is not replicated here because it requires the `MCPanel` R package, for which no Python equivalent exists (see Section 5).

Standard errors. All SEs use the placebo method: randomly designate one control state as “pseudo-treated,” re-run the estimator, and repeat 200 times. The standard deviation of those 200 placebo estimates is the reported SE.

4 Replication Results

Table 1: Effect of Proposition 99 on California Per-Capita Cigarette Sales

Method	Replication	Paper	Std. Error
Synthetic DiD (SDID)	−15.6	−15.6	(8.4)
Synthetic Control (SC)	−19.5	−19.6	(9.7)
Difference-in-Differences (DID)	−27.3	−27.3	(15.3)
De-meaned SC (DIFP)	−11.1	−11.1	(9.3)
Matrix Completion (MC)	—	−20.2	(11.5)

Notes: Outcome is per-capita cigarette sales in packs per year.

Sample: 39 U.S. states, 1970–2000. Treatment: California post-1989.

Standard errors via placebo method (200 replications).

MC not replicated: depends on MCPanel R package (see text).

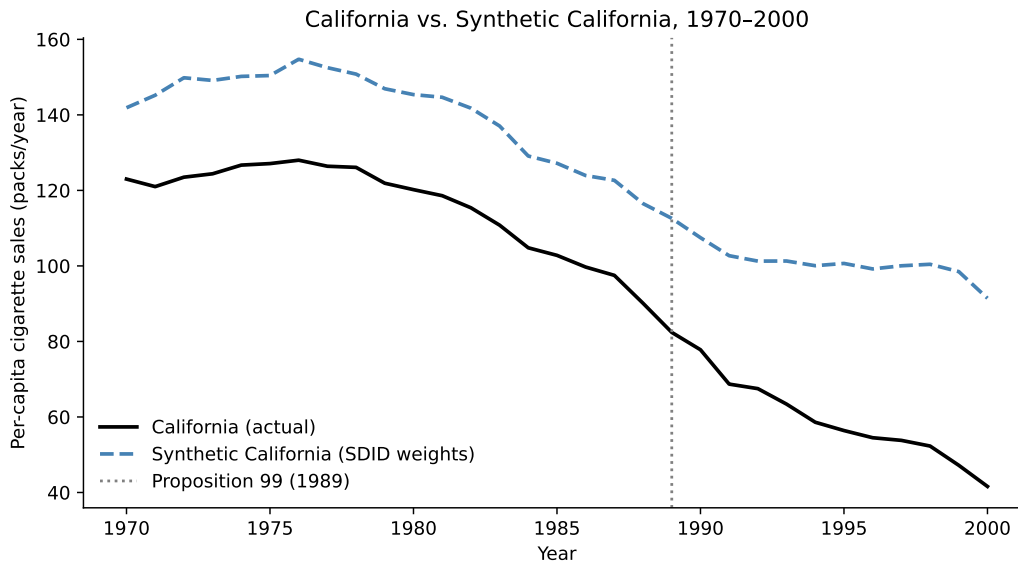


Figure 1: Actual and synthetic California per-capita cigarette sales, 1970–2000. The synthetic series is the SDID-weighted average of the 38 control states. The dotted line marks Proposition 99 (1989).

SDID, SC, DID, and DIFP all match the paper’s Table 1 closely. The figure shows why SDID works well here: the synthetic California tracks actual California almost exactly

through 1988, then diverges sharply after Proposition 99 and that gap is the estimated treatment effect.

5 Challenges in Replication

No official Python package. The authors released only an R package . Two third-party Python ports exist on github but neither works reliably. The PyPI package `synthdid` fails to install on Python 3.13 due to a hard dependency on numpy 1.23.5 as the author’s last updates were 4 years ago. The `pysynthdid` package installs but uses Bayesian optimization to choose the regularization parameter ζ_ω , whereas the paper uses the formula. That difference alone shifts the SDID estimate from -15.6 to approximately -19.8 . Reproducing the paper’s result therefore required translating the authors’ R source code which is located in replication files (`R/solver.R` and `R/synthdid.R`) into NumPy directly with the help of Claude.

MC not replicable in Python. The paper’s MC estimate relies on another R package, which performs cross-validated nuclear norm minimization with two-way fixed effects. No Python equivalent exists, and since the MC isn’t the main result of this paper and would take a similar lengthy process as SDID to work, I left it out.

Slow standard errors. The placebo procedure re-runs full Frank-Wolfe optimization 200 times per estimator, taking about 30 seconds to run.

6 What I Learned

The math translated cleanly once I understood the two key building blocks: the collapsed-form matrix (which reduces the full panel to a compact summary before optimization) and the Frank-Wolfe step (which iteratively shifts weight toward the most useful state or year).

The bigger challenge was practical: the paper excludes python users entirely and leaves them to use third-party packages that either do not install or produce different results.

References

Abadie, A., Diamond, A., and Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of California’s tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505.

- Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., and Wager, S. (2021). Synthetic difference-in-differences. *American Economic Review*, 111(12):4088–4118.
- Athey, S., Bayati, M., Doudchenko, N., Imbens, G., and Khosravi, K. (2021). Matrix completion methods for causal panel data models. *Journal of the American Statistical Association*, 116(536):1716–1730.
- Doudchenko, N. and Imbens, G. W. (2016). Balancing, regression, difference-in-differences and synthetic control methods: A synthesis. Working Paper 22791, National Bureau of Economic Research.